

Global regularity for the Q1-augmented Navier-Stokes system

Track A - this PDE only

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Abstract

On the three-torus, the Navier–Stokes equations with an extra Ladyzhenskaya / p -Laplacian stress (Q1-augmentation, $\varepsilon > 0$, $\beta \geq 1/2$) admit a unique global smooth solution. This is Theorem A on the Domain Architect desk. It is a theorem about this equation. It is not classical three-dimensional Navier–Stokes. The extra dissipation may be sent to zero only after a separate uniform H^1 bound, which remains open; even then classical regularity is a further argument (Track B). The chain records a standard class (Ladyzhenskaya 1968; Málek–Nečas–Růžička) in Q1 notation.

1. The system

Let $\nu > 0$, $\varepsilon > 0$, $\alpha > 0$, and $\beta \geq 1/2$. On $T^3 \times [0, \infty)$ consider

$$\text{dt } u + (u \cdot \text{grad})u = -\text{grad } p + \nu \Delta u + \varepsilon^\alpha P \text{div}(|\text{grad } u|^\beta \text{grad } u), \quad \text{div } u = 0,$$

with u_0 in $H^1(T^3)$ divergence-free. Here P is the Leray projector. Write $p = \beta + 2$, so $\beta \geq 1/2$ is $p \geq 5/2$. The extra term is the variational derivative of $(1/p) \int |\text{grad } u|^p$. It is not the scalar $-\varepsilon^\alpha |\text{grad } u|^\beta \Delta u$.

Theorem A.

Let $\nu > 0$, $\varepsilon > 0$, $\alpha > 0$, $\beta \geq 1/2$, and let u_0 in $H^1(T^3)$ be divergence-free. The Q1 system has a unique solution

$$u \text{ in } C^\infty(T^3 \times (0, \infty)) \cap L^\infty(0, \infty; H^1).$$

No finite-time singularity occurs for this PDE. The data need not be axisymmetric.

2. Proof

Lemma 1 (Energy).

For a smooth solution,

$$\frac{1}{2} \frac{d}{dt} \|u\|_2^2 + \nu \|\text{grad } u\|_2^2 + \varepsilon^\alpha \|\text{grad } u\|_{L^{\beta+2}}^{\beta+2} = 0.$$

Hence for every $T > 0$,

$$\frac{1}{2} \|u(T)\|_2^2 + \nu \int \|\text{grad } u\|_2^2 dt + \varepsilon^\alpha \int \|\text{grad } u\|_{L^{\beta+2}}^{\beta+2} dt = \frac{1}{2} \|u_0\|_2^2.$$

Proof. Test the equation against u . The convective term and the pressure vanish: $\int (u \cdot \text{grad})u \cdot u = \frac{1}{2} \int u \cdot \text{grad}(|u|^2) = 0$. The extra stress gives $\int P \text{div}(|\text{grad } u|^\beta \text{grad } u) \cdot u = -\int |\text{grad } u|^{\beta+2}$. The Stokes term is $-\nu \|\text{grad } u\|_2^2$.

Lemma 2 (Galerkin).

The Galerkin projections onto the first n Stokes eigenfunctions exist globally. The energy bounds are uniform in n . A subsequence converges weakly to a weak solution

$$u \text{ in } L^\infty(0, \infty; L^2) \cap L^2(0, \infty; H^1), \quad \text{grad } u \text{ in } L^{\beta+2}(0, \infty; L^{\beta+2}).$$

Proof. Finite-dimensional ODE; the same energy identity holds. No finite-time blowup of $\|u_n\|_2$.

Banach–Alaoglu and Aubin–Lions pass to the limit. Monotonicity of $v \mapsto \operatorname{div}(|\operatorname{grad} v|^\beta \operatorname{grad} v)$ is Minty–Browder. Details: Málek–Nečas–Růžička, Weak and Measure-valued Solutions to Evolutionary PDEs, Chapter 5.

Lemma 3 (Strong solutions, $\beta \geq 1/2$).

The weak solution is unique and lies in $L^\infty(0, \infty; H^1) \cap L^2(0, \infty; H^2)$. The constant depends on $\varepsilon, \alpha, \beta, v, \|u_0\|_{H^1}$ and blows up as $\varepsilon \rightarrow 0$.

Proof. For $\beta \geq 1/2$ one has $\beta + 2 \geq 5/2$. Extra integrability of $\operatorname{grad} u$ meets the Ladyzhenskaya $p \geq 5/2$ criterion (equivalently a Ladyzhenskaya–Prodi–Serrin pair built from that integrability). The extra term produces a monotone remainder that absorbs the convective difference of two solutions. See Ladyzhenskaya (1968) and Málek–Nečas–Růžička. This is extra integrability of $\operatorname{grad} u$, not a geometric cancel, and not a Φ identity.

Lemma 4 (Bootstrap).

The unique strong solution is $C^\infty(T^3 \times (0, \infty))$.

Proof. Frozen $\varepsilon > 0$, the linearized operator is a uniformly elliptic Stokes operator. Difference quotients give H^k for all k . Sobolev embedding yields spatial smoothness; time regularity follows from the equation.

Proof of Theorem A.

Galerkin existence, weak limit, strong uniqueness at $\beta \geq 1/2$, then the bootstrap.

3. What this does not prove

Uniform H^1 as $\varepsilon \rightarrow 0$ is not claimed. A decaying Q1 integral is not that bound. Classical unaugmented Navier–Stokes (Track B: keep $1/r^4$, no Q1) is not claimed. $A \Rightarrow B$ remains false on this desk. Riemann zeros, Birch–Swinnerton-Dyer, Hodge, Yang–Mills, and P vs NP are different objects.

References

- [1] O. A. Ladyzhenskaya, The Mathematical Theory of Viscous Incompressible Flow, Gordon and Breach, 1969 (modified NS: extra p -Laplacian stress).
- [2] J. Málek, J. Nečas, M. Růžička, Weak and Measure-valued Solutions to Evolutionary PDEs, Chapman & Hall, 1996.